From the Quantum Phase estimation algorithm to the HHL and QAOA algorithms

Morten Hjorth-Jensen

FYS5419/9419: Week of April 7-11, 2026

Outline

1. Quantum Phase Estimation (QPE)
2. Connection to Hamiltonian simulation
3. Connection to Variational Quantum Eigensolvers (VQE)
4. Introducing the HHL algorithm

See also separate jupyter-notebooks at `https://github.com/CompPhysics/QuantumComputingMachineLearning/tree/gh-pages/doc/pub/week11/ipynb/week11.ipynb`

Quantum Phase Estimation: Problem Statement, reminder from Lecture March 25

Suppose U is a unitary operator and $|\psi\rangle$ is one of its eigenstates:

$$U|\psi\rangle = e^{2\pi i\phi}|\psi\rangle, \qquad 0 \leq \phi < 1.$$

The goal of Quantum Phase Estimation (QPE) is to estimate the phase ϕ.
Equivalently, QPE extracts the eigenvalue of U in phase form.

Why QPE Matters

QPE is central because many physical problems can be mapped to eigenvalue estimation.
Examples:

- energy estimation in quantum chemistry,
- eigenvalue problems in many-body physics,
- Hamiltonian simulation,
- Shor's algorithm and period finding.

If we can encode a Hamiltonian into a unitary, then QPE can extract its eigenvalues.

Registers Used in QPE

QPE uses two registers:

- a control register of n qubits, initialized in $|0\rangle^{\otimes n}$,
- a system register, initialized in the eigenstate $|\psi\rangle$.

So the initial state is

$$|0\rangle^{\otimes n} |\psi\rangle .$$

The control register stores the estimated phase bits.

Step 1: Create a Uniform Superposition

Apply Hadamard gates to all control qubits:

$$|0\rangle^{\otimes n} \mapsto \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} |k\rangle\,.$$

So the total state becomes

$$\frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} |k\rangle\,|\psi\rangle\,.$$

This prepares the control register as a coherent superposition of all binary numbers k.

Step 2: Controlled Powers of U

Now apply controlled powers of U:

$$U^{2^0}, U^{2^1}, U^{2^2}, \ldots, U^{2^{n-1}}.$$

Each control qubit determines whether the corresponding power is applied.
Because $|\psi\rangle$ is an eigenstate,

$$U^{2^j} |\psi\rangle = e^{2\pi i 2^j \phi} |\psi\rangle .$$

Hence each controlled unitary writes phase information into the control register.

Step 2: State After Controlled Powers

If the control register basis state is $|k\rangle$, then the controlled operations apply U^k, giving

$$U^k |\psi\rangle = e^{2\pi i k\phi} |\psi\rangle .$$

Therefore the combined state becomes

$$\frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i k\phi} |k\rangle |\psi\rangle .$$

The system register factors out, and the control register now contains the phase information.

Key Observation

The control register is in the state

$$\frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i k \phi} |k\rangle ,$$

which is exactly a Fourier-type state.
This is why the inverse QFT is the correct next step: it converts this phase-encoded superposition into a computational basis approximation of ϕ.

Step 3: Apply the Inverse QFT

Applying QFT^{-1} to the control register gives

$$\mathrm{QFT}^{-1}\left(\frac{1}{2^{n/2}}\sum_{k=0}^{2^n-1} e^{2\pi i k\phi}\,|k\rangle\right).$$

If

$$\phi = \frac{m}{2^n}$$

exactly for some integer m, then this becomes

$$|m\rangle\,.$$

So the control register stores the exact n-bit binary expansion of ϕ.

Exact QPE Case

Suppose

$$\phi = 0.\phi_1\phi_2\ldots\phi_n = \frac{m}{2^n}.$$

Then after inverse QFT, the control register becomes

$$|\phi_1\phi_2\ldots\phi_n\rangle .$$

Thus measuring the control register returns the exact binary digits of ϕ.
This is the cleanest case of QPE.

Step-by-Step Derivation with Binary Fractions

Let

$$\phi = 0.\phi_1\phi_2 \ldots \phi_n \ldots$$

Then one can show that the control-register state after controlled powers factorizes as

$$\frac{1}{2^{n/2}} \left(|0\rangle + e^{2\pi i\, 2^{n-1}\phi} |1\rangle\right) \otimes \left(|0\rangle + e^{2\pi i\, 2^{n-2}\phi} |1\rangle\right) \otimes \cdots$$

$$\cdots \otimes \left(|0\rangle + e^{2\pi i\, \phi} |1\rangle\right).$$

Because

$$2^{n-1}\phi = \phi_1.\phi_2\phi_3 \ldots,$$

only the fractional parts matter in the phases, so each qubit encodes one successive binary suffix.

How the Bits Emerge

Using the binary expansion of ϕ, the control-register state becomes

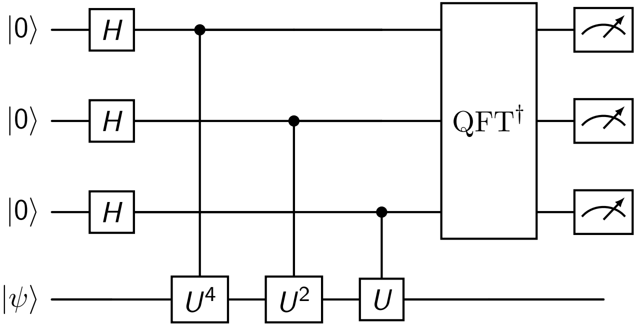
$$\frac{1}{2^{n/2}}\left(|0\rangle + e^{2\pi i\, 0.\phi_n}|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i\, 0.\phi_{n-1}\phi_n}|1\rangle\right)\otimes\cdots$$

$$\cdots\otimes\left(|0\rangle + e^{2\pi i\, 0.\phi_1\phi_2\ldots\phi_n}|1\rangle\right),$$

which has exactly the structure of a QFT state.
Therefore applying QFT^{-1} yields the bit string

$$\phi_1\phi_2\ldots\phi_n.$$

QPE Circuit



This example shows a 3-qubit control register and one system register.

Worked Example: $\phi = 5/8$

Suppose

$$\phi = \frac{5}{8} = 0.101.$$

Then the ideal QPE output on a 3-qubit control register is

$$|101\rangle .$$

Indeed,

$$e^{2\pi i\phi} = e^{2\pi i(5/8)}.$$

After the controlled powers and inverse QFT, the control register collapses to the bit string 101 with probability 1.

When ϕ Is Not Exactly Representable

If ϕ is not exactly of the form $m/2^n$, then QPE does not return one exact bit string.
Instead, the output is a probability distribution peaked near the best n-bit approximation:

$$\phi \approx \frac{m}{2^n}.$$

The measurement outcome gives the nearest binary approximation with high probability.

Precision of QPE

With n control qubits, QPE resolves phases to approximately

$$\Delta\phi \sim \frac{1}{2^n}.$$

Thus the precision improves exponentially with the number of control qubits.
This exponential precision is one of the main reasons QPE is so powerful.

From Phases to Energies

Suppose H is a Hamiltonian with eigenstate $|E\rangle$:

$$H|E\rangle = E|E\rangle.$$

Define the time-evolution operator

$$U(t) = e^{-iHt}.$$

Then

$$U(t)|E\rangle = e^{-iEt}|E\rangle.$$

Comparing with

$$U|\psi\rangle = e^{2\pi i\phi}|\psi\rangle,$$

we identify

$$\phi = -\frac{Et}{2\pi} \mod 1.$$

Thus QPE can estimate energies if we can implement e^{-iHt}.

Connection to Hamiltonian Simulation

QPE requires controlled applications of

$$U^{2^j} = e^{-iHt2^j}.$$

So QPE depends directly on the ability to perform Hamiltonian simulation.
In practice this may be done using:

- Trotter-Suzuki decompositions,
- linear-combination-of-unitaries methods,
- qubitization,
- problem-specific simulation techniques.

Thus:

$$\text{Hamiltonian simulation} + \text{QPE} \quad \Rightarrow \quad \text{energy estimation.}$$

Physical Interpretation

QPE measures the dynamical phase accumulated by an energy eigenstate under time evolution.
The algorithm turns the phase

$$e^{-iEt}$$

into a bit string encoding the energy.
So one can view QPE as a highly precise interferometric energy-measurement protocol.

Why QPE Is Important in Quantum Chemistry

In quantum chemistry and many-body physics, the main goal is often to compute eigenenergies of a Hamiltonian.
If we can prepare an approximate eigenstate with sufficient overlap with the true ground state, then QPE can project onto the exact eigenvalue with high precision.
This makes QPE a natural route to:

- ground-state energies,
- excited-state energies,
- spectral properties.

Connection to VQE

The Variational Quantum Eigensolver (VQE) and QPE aim at similar physics goals, but in different ways.

VQE:

- variational,
- hybrid quantum-classical,
- optimized by minimizing

$$E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})| H |\psi(\boldsymbol{\theta})\rangle ,$$

- suitable for noisy near-term devices.

QPE:

- not variational,
- requires coherent controlled time evolution,
- provides direct eigenvalue estimation,
- suited to fault-tolerant quantum computers.

How VQE and QPE Complement Each Other

A common long-term strategy is:

1. Use a variational method such as VQE to prepare a good approximate eigenstate
$$|\psi_{\text{trial}}\rangle$$
with large overlap with the true eigenstate.
2. Use that state as the input to QPE.
3. QPE then refines the energy estimate to high precision.

So VQE can be viewed as a state-preparation front end for QPE.

Overlap Requirement

If the initial state is

$$|\psi_{\mathrm{trial}}\rangle = \sum_j c_j |E_j\rangle ,$$

then QPE outputs E_j with probability

$$|c_j|^2.$$

Therefore the success probability depends on the overlap with the desired eigenstate.
This is exactly why improved ansätze in VQE are also valuable for QPE-based workflows.

QPE vs VQE in Practice

VQE

- shallow circuits,
- many repeated measurements,
- classical optimization loop,
- noise tolerant,
- limited by optimization difficulty and shot cost.

QPE

- deep coherent circuits,
- requires controlled e^{-iHt},
- very precise,
- ideal for fault-tolerant hardware,
- often viewed as the long-term gold standard for energy estimation.

Semiclassical Interpretation

One may interpret QPE as repeatedly extracting successive binary digits of a phase.

The inverse QFT acts globally, but conceptually it decodes the phase one bit at a time.

This bitwise structure is one reason QPE is so elegant mathematically.

Implementation

A implementation of the QPE typically proceeds as follows:

1. choose the number of control qubits,
2. initialize the system register in an eigenstate or approximate eigenstate,
3. apply Hadamards to the control register,
4. apply controlled U^{2^j},
5. apply inverse QFT,
6. measure the control register,
7. interpret the measurement as an estimate of ϕ.

If the notebook also implements the QFT explicitly, it will usually reuse the same Hadamard + controlled-phase + swap pattern.

Summary of the Full Picture

$$\text{Hamiltonian } H \quad \longrightarrow \quad U(t) = e^{-iHt} \quad \longrightarrow \quad \text{QPE} \quad \longrightarrow \quad E$$

The QFT is the phase-decoding engine inside QPE.
VQE and QPE are complementary:

- VQE is a near-term variational energy estimator,
- QPE is a long-term high-precision eigenvalue estimator,
- VQE can provide input states for QPE.

Summary

We have seen that:

- the QFT maps

$$|x\rangle \mapsto \frac{1}{\sqrt{N}} \sum_k e^{2\pi i x k / N} |k\rangle ,$$

- QPE uses controlled powers of a unitary and the inverse QFT to extract eigenphases,
- for Hamiltonian evolution

$$U(t) = e^{-iHt},$$

  QPE becomes an energy-estimation algorithm,
- VQE and QPE address related eigenvalue problems with different hardware requirements.

Suggestions for the second project

- Continuation of project 1 with more realistic systems and using adaptive VQE on actual (real) quantum computers
- Implementation and studies of the Quantum Approximate Optimization Algorithm (QAOA): Simulation of linear algebra systems on quantum computers, solving for example differential equations
- Implementation of the HHL algorithm (needs QPE)
- Applications and implementations of quantum machine learning algorithms
- Implement quantum neural networks for PINNs
- Detailed quantum chemistry applications.

Suggestions for the second project

- Studies of entanglement and physical realization of quantum gates and circuits
- Implementing Shor's algorithm or other algorithms
- Parallelizing the Variational Quantum Eigensolver with multi-GPU scaling, see `https://arxiv.org/abs/2601.09951`
- Implementing and comparing the quantum phase estimation algorithm with VQE, see for example recent QPE paper at `"https://arxiv.org/abs/2601.05788`
- Studies of hyperparameters in VQE, see for example `https://arxiv.org/abs/2601.16679`
- Implementing unitary coupled cluster theory on a quantum computer, see `https://arxiv.org/abs/2501.15893`

What is the HHL algorithm?

The HHL algorithm—named after Aram Harrow, Avinatan Hassidim, and Seth Lloyd—is one of the central quantum algorithms for solving linear systems of equations.
The core idea is to solve a linear system

$$A\mathbf{x} = \mathbf{b}.$$

What is the QAOA algorithm?, to be discussed April 15

QAOA stands for the Quantum Approximate Optimization Algorithm

Each part of the name reflects something essential about the method:

- Quantum: it runs on a quantum computer, using qubits and quantum gates.
- Approximate: it typically finds near-optimal solutions rather than exact ones.
- Optimization: it is designed to solve optimization problems (like MaxCut, scheduling, portfolio selection).
- Algorithm: it is a well-defined hybrid quantum–classical procedure.

QAOA is a variational quantum algorithm that uses a parameterized quantum circuit to search for good solutions to hard optimization problems.

HHL: Linear Systems Everywhere

- Solve:

$$A\mathbf{x} = \mathbf{b}$$

- Appears in:
  - PDEs
  - Machine learning
  - Many-body physics
- Classical cost: $\mathcal{O}(N^3)$ (dense case)

Quantum Goal

- Encode vector as quantum state:

$$|b\rangle = \sum_i b_i |i\rangle$$

- Goal:

$$|x\rangle \propto A^{-1}|b\rangle$$

- Output is a quantum state, not a classical vector

Eigenbasis Representation

$$A|u_j\rangle = \lambda_j|u_j\rangle$$
$$|b\rangle = \sum_j \beta_j|u_j\rangle$$

Formal Solution

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j}|u_j\rangle$$

- Problem reduces to eigenvalue inversion

Hamiltonian Encoding

- Assume A is Hermitian
- Implement:

$$U = e^{iAt}$$

QPE Action

$$|u_j\rangle \to |u_j\rangle|\lambda_j\rangle$$

State After QPE

$$\sum_j \beta_j |u_j\rangle |\lambda_j\rangle$$

Key Step: Inversion

- Introduce ancilla qubit
- Perform controlled rotation:

$$|\lambda_j\rangle \to \sqrt{1 - \frac{C^2}{\lambda_j^2}}|0\rangle + \frac{C}{\lambda_j}|1\rangle$$

State After Rotation

$$\sum_j \beta_j |u_j\rangle |\lambda_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}} |0\rangle + \frac{C}{\lambda_j} |1\rangle \right)$$

Inverse QPE

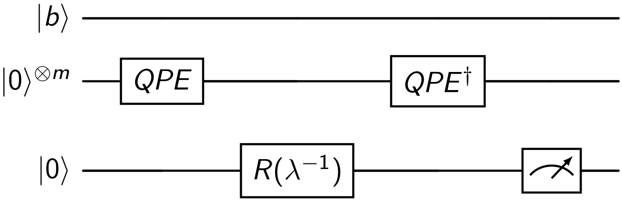
- Remove eigenvalue register
- Leaves:

$$\sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

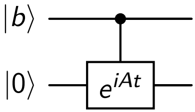
Final State

$$|x\rangle \propto A^{-1}|b\rangle$$

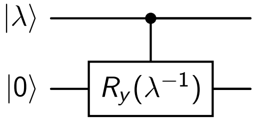
HHL Circuit (Overview)



Detailed Circuit Step 1: QPE



Detailed Circuit Step 2: Controlled Rotation



Detailed Circuit Step 3: Uncomputation

Post-selection and measurement

- Measure ancilla
- Condition on $|1\rangle$
- Success probability depends on κ

Runtime

$$\mathcal{O}\left(\kappa^2 \log N\right)$$

- κ = condition number

Conditions for Speedup

- Sparse matrix
- Efficient Hamiltonian simulation
- Efficient state preparation

Readout Problem

- Cannot extract full vector efficiently
- Only expectation values accessible

Condition Number Issue

- Runtime scales with κ
- Ill-conditioned systems are hard

Operator View

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle\langle u_j|$$

Connection to Green's Functions

$$G = (A)^{-1}$$

- Resolvent operator
- Linear response theory

Many-Body Perspective

- HHL = spectral filtering
- Similar to:
  - Imaginary time evolution
  - Projection methods

Applications

- Quantum machine learning
- PDE solvers
- Data fitting

Summary HHL

- HHL solves $Ax = b$ in quantum form
- Uses:
  - Phase estimation
  - Controlled rotation
  - Uncomputation
- Connects strongly to spectral physics

From Linear Systems to Resolvents, HHL and Green's function theory

A central observation is that HHL solves a linear system

$$Ax = b$$

by preparing a quantum state proportional to

$$|x\rangle \propto A^{-1}|b\rangle.$$

In many-body physics, the inverse of an operator is precisely the basic structure of a *resolvent* or *Green's function*.
If we define

$$G(z) = (zI - H)^{-1},$$

then solving

$$(zI - H)|x\rangle = |b\rangle$$

is equivalent to preparing

$$|x\rangle = G(z)|b\rangle.$$

Thus HHL may be viewed as a quantum algorithm for applying a resolvent operator to a source state.

Spectral Form of the Green's Function

Let

$$H|u_j\rangle = E_j|u_j\rangle.$$

Then the resolvent has the spectral decomposition

$$G(z) = (zI - H)^{-1} = \sum_j \frac{1}{z - E_j}|u_j\rangle\langle u_j|.$$

If

$$|b\rangle = \sum_j \beta_j|u_j\rangle,$$

then

$$G(z)|b\rangle = \sum_j \frac{\beta_j}{z - E_j}|u_j\rangle.$$

This is structurally identical to the HHL transformation:

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j}|u_j\rangle.$$

Hence, HHL implements the same spectral inversion mechanism that underlies Green's functions.

Physical Meaning of the Spectral Filter

The factor

$$\frac{1}{z - E_j}$$

acts as a spectral filter.

- Eigenmodes with energies near z are enhanced.
- Eigenmodes far from z are suppressed.
- The inverse operator encodes propagation and response properties.

In this sense, HHL is not merely a linear solver. It is a *spectral transformation algorithm*: it maps a source state to a response state through an inverse operator.

Retarded Green's Function

In condensed matter and many-body theory, one often uses

$$G^R(\omega) = \frac{1}{\omega + i\eta - H}, \qquad \eta > 0.$$

The small positive η regularizes the inverse and enforces causal boundary conditions.

Applying this operator to a source state gives

$$|x(\omega)\rangle = G^R(\omega)|b\rangle.$$

Formally, an HHL-type algorithm can be interpreted as a way of preparing a quantum state proportional to this frequency-dependent response vector, provided one can encode the shifted operator

$$A(\omega) = \omega I - H$$

or

$$A(\omega) = \omega I + i\eta - H.$$

Connection to Correlation Functions

Many observables in quantum many-body physics can be written in terms of Green's functions or resolvents.
For an operator $\hat{O}$, a typical linear-response quantity has the form

$$\chi(\omega) = \langle \psi_0 | \hat{O}^\dagger \frac{1}{\omega + i\eta - (H - E_0)} \hat{O} | \psi_0 \rangle.$$

If we define the source state

$$|b\rangle = \hat{O} |\psi_0\rangle,$$

then

$$\chi(\omega) = \langle b | \frac{1}{\omega + i\eta - (H - E_0)} | b \rangle.$$

This shows that HHL-like inversion is directly connected to computing dynamical susceptibilities.

HHL as a Green's Function Engine

At a conceptual level, HHL performs three tasks:

1. Decompose the source state into eigenmodes.
2. Invert the spectral coefficients:

$$\lambda_j \mapsto \frac{1}{\lambda_j}.$$

3. Recombine the filtered eigenmodes into the response state.

This is exactly what is needed for:

- resolvent methods,
- Green's function methods,
- frequency-domain linear response,
- projection-based many-body calculations.

Linear Response in Physics

Suppose a system with Hamiltonian H_0 is perturbed by

$$V(t) = -f(t)\hat{F}.$$

To first order, the expectation value of an observable $\hat{O}$ changes as

$$\delta\langle \hat{O}(t)\rangle = \int dt'\, \chi_{OF}(t-t')f(t'),$$

where χ_{OF} is the response function.
In the frequency domain, one typically writes

$$\delta\langle \hat{O}(\omega)\rangle = \chi_{OF}(\omega)f(\omega).$$

The response function is often expressible through inverse operators of the form

$$(\omega I - M)^{-1},$$

which is precisely the structure targeted by HHL.

Resolvent Form of the Susceptibility

A standard spectral representation is

$$\chi_{OF}(\omega) = \sum_{n\neq 0} \left(\frac{\langle 0|\hat{O}|n\rangle\langle n|\hat{F}|0\rangle}{\omega - (E_n - E_0) + i\eta} - \frac{\langle 0|\hat{F}|n\rangle\langle n|\hat{O}|0\rangle}{\omega + (E_n - E_0) + i\eta} \right).$$

This is a sum over poles, or equivalently a matrix element of a resolvent.

Thus linear response is fundamentally an *inverse-operator problem*, which places it in the same mathematical class as HHL.

What is Time-Dependent Hartree–Fock?

Time-dependent Hartree–Fock (TDHF) describes the evolution of a Slater determinant under a time-dependent mean field.
At the one-body density-matrix level,

$$i\hbar \frac{d\rho}{dt} = [h[\rho], \rho].$$

To study small oscillations around a stationary solution ρ_0, one linearizes:

$$\rho(t) = \rho_0 + \delta\rho(t),$$

leading to an equation of motion for the fluctuation $\delta\rho$.

Linearized TDHF

After linearization, one obtains a matrix equation of the form

$$i\hbar\frac{d}{dt}\delta\rho = \mathcal{L}\,\delta\rho + s(t),$$

where:

- $\mathcal{L}$ is the linearized Liouvillian or RPA matrix,
- $s(t)$ is an external driving term.

In the frequency domain, this becomes

$$(\omega I - \mathcal{L})\delta\rho(\omega) = s(\omega).$$

This is now explicitly a linear system. Its solution is

$$\delta\rho(\omega) = (\omega I - \mathcal{L})^{-1} s(\omega).$$

That is exactly the algebraic structure addressed by HHL.

TDHF as an HHL-Type Problem

The formal similarity is immediate:

$$Ax = b \qquad \Longleftrightarrow \qquad (\omega I - \mathcal{L})\delta\rho(\omega) = s(\omega).$$

Hence:

$$A \leftrightarrow \omega I - \mathcal{L}, \qquad x \leftrightarrow \delta\rho(\omega), \qquad b \leftrightarrow s(\omega).$$

If one can encode the linear-response matrix $\mathcal{L}$ into a quantum representation, then an HHL-type algorithm could in principle prepare a state proportional to the TDHF response.

Connection to Random Phase Approximation (RPA)

The small-amplitude limit of TDHF is closely related to the Random Phase Approximation (RPA).
One typically obtains a non-Hermitian eigenvalue problem of the form

$$\begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \omega \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Equivalently, one can formulate a driven linear-response problem

$$(\omega I - M)\, v = s.$$

Again, this is an inverse problem. The response amplitudes are obtained by applying the resolvent

$$(\omega I - M)^{-1}$$

to a source vector.

What HHL Would Compute in TDHF

In a TDHF or RPA context, one is often interested in:

- transition amplitudes,
- density oscillations,
- response to an external field,
- strength functions.

These quantities can often be expressed as matrix elements involving the response vector:

$$S(\omega) \sim \langle s|(\omega I - M)^{-1}|s\rangle.$$

Thus an HHL-type routine would not need to output the full vector $\delta\rho(\omega)$. It would already be useful if it allowed efficient estimation of such expectation values.

Strength Functions and Dynamical Response

A common observable in TDHF and RPA is the strength function

$$S_F(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,\chi_F(\omega),$$

where

$$\chi_F(\omega) = \langle 0|\hat{F}^\dagger \frac{1}{\omega + i\eta - (H - E_0)} \hat{F}|0\rangle.$$

This is exactly a Green's function expression. Hence:

TDHF/RPA response $\longleftrightarrow$ resolvent evaluation $\longleftrightarrow$ HHL-type i

Why This Connection is Interesting

The HHL framework is potentially relevant because:

- linear response is naturally formulated as a linear system,
- inverse operators define propagators and susceptibilities,
- many observables are expectation values rather than full vectors,
- this aligns well with quantum output models.

From a physics viewpoint, HHL is best interpreted not only as a solver of $Ax = b$, but as a quantum method for preparing *response states* and probing *spectral properties*.

Important Caveats

There are, however, important obstacles:

- The matrix A must be efficiently block-encoded or simulated.
- TDHF/RPA matrices may be non-Hermitian.
- Condition numbers may be large near resonances.
- State preparation and measurement may dominate the cost.

Therefore, the connection is mathematically deep and physically appealing, but practical advantage depends strongly on encoding, sparsity, conditioning, and the observables of interest.

Hermitian Embedding of Non-Hermitian Problems

If the response matrix M is non-Hermitian, one common strategy is Hermitian embedding:

$$\mathcal{H} = \begin{pmatrix} 0 & M \\ M^\dagger & 0 \end{pmatrix}.$$

Then $\mathcal{H}$ is Hermitian, and one may reformulate the problem in an enlarged space.

This is often the natural bridge between:

- non-Hermitian response equations,
- quantum phase estimation or HHL-type methods,
- Green's function formulations on quantum hardware.

Conceptual Summary

Key idea

HHL applies an inverse operator spectrally:

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle\langle u_j|.$$

Green's function viewpoint

Choose

$$A = zI - H,$$

then HHL prepares a state proportional to

$$(zI - H)^{-1}|b\rangle.$$

TDHF / linear-response viewpoint

Choose

$$A = \omega I - \mathcal{L},$$

Linear Systems in Physics

Many physics problems reduce to

$$Ax = b.$$

Examples:

- discretized Schrödinger and diffusion equations,
- linear response theory,
- Green's function methods,
- time-dependent Hartree–Fock (TDHF),
- random phase approximation (RPA).

The HHL algorithm prepares a quantum state proportional to

$$|x\rangle \propto A^{-1}|b\rangle.$$

The Core Spectral Problem

Suppose

$$A|u_j\rangle = \lambda_j|u_j\rangle, \qquad |b\rangle = \sum_j \beta_j|u_j\rangle.$$

Then

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j}|u_j\rangle.$$

Thus HHL implements the spectral map

$$\lambda_j \mapsto \frac{1}{\lambda_j}.$$

This is why HHL is best viewed as a *spectral inverse-operator algorithm*.

Spectral Decomposition of the Operator

Write

$$A = \sum_j \lambda_j |u_j\rangle\langle u_j|.$$

Then formally

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle\langle u_j|.$$

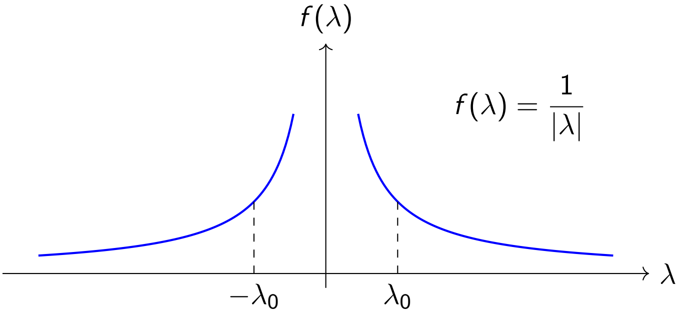
Applying the inverse to the source state:

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle.$$

The algorithm must therefore:

1. identify the spectral components,
2. apply the inverse weight $1/\lambda_j$,
3. recombine the result into a state.

Spectral Filter View



HHL applies a *spectral filter*:

$$f(\lambda) = \frac{1}{\lambda}.$$

Modes with small $|\lambda|$ are enhanced, while large-$|\lambda|$ modes are suppressed.

Why Quantum Phase Estimation is Needed

In the original HHL algorithm, we must attach eigenvalue information to each eigenvector:

$$|u_j\rangle|0\rangle \longrightarrow |u_j\rangle|\lambda_j\rangle.$$

This is done by **quantum phase estimation (QPE)**.
After QPE, the state becomes

$$\sum_j \beta_j |u_j\rangle|\lambda_j\rangle,$$

and only then can we implement

$$\lambda_j \mapsto \frac{1}{\lambda_j}$$

through a controlled ancilla rotation.

Setup for QPE

Assume A is Hermitian and can be exponentiated:

$$U = e^{iAt}.$$

If

$$A|u_j\rangle = \lambda_j|u_j\rangle,$$

then

$$U|u_j\rangle = e^{i\lambda_j t}|u_j\rangle.$$

Thus the eigenvalue problem for A becomes a phase problem for U:

$$e^{i\lambda_j t} = e^{2\pi i\phi_j}, \qquad \phi_j = \frac{\lambda_j t}{2\pi}.$$

QPE Registers

QPE uses:

- a *system register* holding the eigenvector,
- a *clock register* of m qubits for phase readout.

Initial state:

$$|u_j\rangle \otimes |0\rangle^{\otimes m}.$$

After Hadamards on the clock register:

$$|u_j\rangle \otimes \frac{1}{2^{m/2}} \sum_{k=0}^{2^m-1} |k\rangle.$$

Controlled Powers of the Unitary

Apply controlled powers of U:

$$U^{2^0}, U^{2^1}, \ldots, U^{2^{m-1}}.$$

Since

$$U^{2^\ell}|u_j\rangle = e^{2\pi i 2^\ell \phi_j}|u_j\rangle,$$

the phase information is kicked back onto the control register. The state becomes

$$\frac{1}{2^{m/2}} \sum_{k=0}^{2^m-1} e^{2\pi i k \phi_j}|k\rangle|u_j\rangle.$$

Inverse Quantum Fourier Transform

Applying the inverse quantum Fourier transform (QFT^{-1}) to the clock register gives an estimate of ϕ_j, and hence of λ_j:

$$|u_j\rangle|0\rangle^{\otimes m} \longrightarrow |u_j\rangle|\widetilde{\lambda}_j\rangle.$$

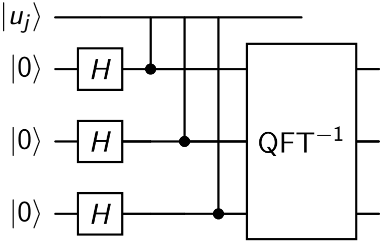
Thus, for a superposition

$$|b\rangle = \sum_j \beta_j |u_j\rangle,$$

QPE yields

$$\sum_j \beta_j |u_j\rangle|\widetilde{\lambda}_j\rangle.$$

QPE Circuit Block



Here the controlled gates are

$$U, \quad U^2, \quad U^4, \ldots$$

with $U = e^{iAt}$.

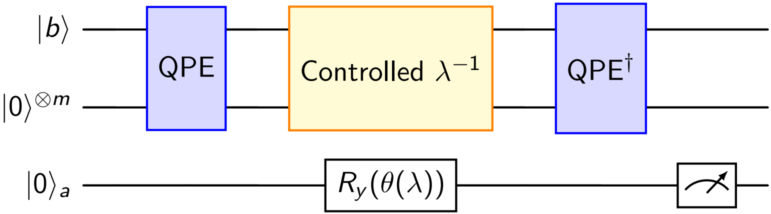
QPE Logic in Words

QPE performs three steps:

1. Create a superposition of time labels in the clock register.
2. Let each branch accumulate a phase $e^{i\lambda_j t}$.
3. Fourier analyze those phases to recover λ_j.

Hence QPE acts as a quantum spectral analyzer.

High-Level HHL Circuit



- Blue block: spectral decomposition and recombination.
- Yellow block: ancilla rotation implementing the inverse eigenvalue weight.

Step 1: Input State

Prepare

$$|b\rangle = \sum_j \beta_j |u_j\rangle.$$

The full initial state is

$$\sum_j \beta_j |u_j\rangle |0\rangle^{\otimes m} |0\rangle_a.$$

Step 2: After QPE

After QPE:

$$\sum_j \beta_j |u_j\rangle |0\rangle^{\otimes m} |0\rangle_a \longrightarrow \sum_j \beta_j |u_j\rangle |\lambda_j\rangle |0\rangle_a.$$

This is the stage where the spectral decomposition becomes explicit.

Step 3: Controlled Rotation

Choose the ancilla rotation so that

$$\sin \frac{\theta(\lambda_j)}{2} = \frac{C}{\lambda_j},$$

for some constant C.
Then

$$|\lambda_j\rangle|0\rangle_a \mapsto |\lambda_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}} |0\rangle_a + \frac{C}{\lambda_j} |1\rangle_a \right).$$

State After Controlled Rotation

The total state becomes

$$\sum_j \beta_j |u_j\rangle |\lambda_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}} |0\rangle_a + \frac{C}{\lambda_j} |1\rangle_a \right).$$

The inverse eigenvalue weights are now encoded in the ancilla amplitudes.

Step 4: Inverse QPE

Apply $\mathrm{QPE}^\dagger$ to uncompute the eigenvalue register:

$$\sum_j \beta_j |u_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}} |0\rangle_a + \frac{C}{\lambda_j} |1\rangle_a \right).$$

Now the eigenvalue register is disentangled, and only the filtered amplitudes remain.

Step 5: Postselection

Conditioning on the ancilla outcome $|1\rangle_a$, the system collapses to

$$|x\rangle \propto \sum_j \frac{\beta_j}{\lambda_j}|u_j\rangle = A^{-1}|b\rangle.$$

This is the desired quantum state solution.

Operator-Function View

HHL implements

$$f(A)|b\rangle$$

with

$$f(\lambda) = \frac{1}{\lambda}.$$

More generally, the logic is:

- diagonalize the operator,
- transform eigenvalues with a desired function,
- reconstruct the state.

This places HHL in the wider class of quantum spectral transformation methods.

Comparison with Time Evolution

Time evolution applies

$$e^{-iAt} = \sum_j e^{-i\lambda_j t} |u_j\rangle\langle u_j|.$$

HHL applies

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle\langle u_j|.$$

So:

- dynamics multiplies modes by phases,
- HHL multiplies modes by inverse eigenvalues.

This is much closer to response theory than to ordinary time evolution.

Resolvent Operator

In many-body physics, define the resolvent

$$G(z) = (zI - H)^{-1}.$$

If

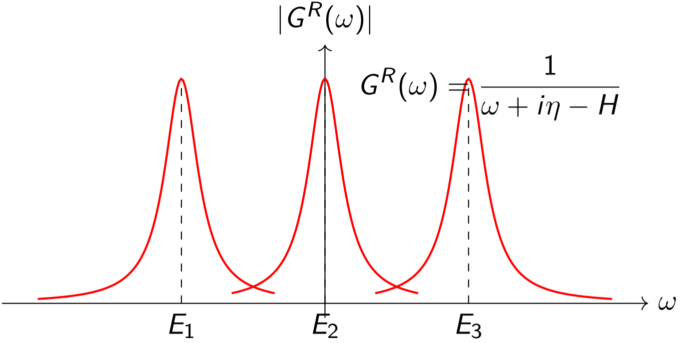
$$H|u_j\rangle = E_j|u_j\rangle,$$

then

$$G(z) = \sum_j \frac{1}{z - E_j} |u_j\rangle\langle u_j|.$$

This is exactly the same spectral structure as HHL.

Green's Function Poles



The poles of the Green's function correspond to the excitation energies of the system.

Applying a Green's Function to a Source State

Let

$$|b\rangle = \sum_j \beta_j |u_j\rangle.$$

Then

$$G(z)|b\rangle = \sum_j \frac{\beta_j}{z - E_j} |u_j\rangle.$$

This has exactly the same structure as

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle.$$

Thus HHL can be interpreted as a quantum Green's-function-style filtering procedure.

Dynamical Susceptibility

A standard linear-response object is

$$\chi(\omega) = \langle \psi_0 | \hat{O}^\dagger \frac{1}{\omega + i\eta - (H - E_0)} \hat{O} | \psi_0 \rangle.$$

Defining the source

$$|b\rangle = \hat{O}|\psi_0\rangle,$$

we obtain

$$\chi(\omega) = \langle b | \frac{1}{\omega + i\eta - (H - E_0)} | b \rangle.$$

Hence HHL is naturally connected to the preparation of response states.

TDHF Equation

Time-dependent Hartree–Fock is governed by

$$i\frac{d\rho}{dt} = [h[\rho], \rho].$$

Linearizing about a stationary solution,

$$\rho(t) = \rho_0 + \delta\rho(t),$$

yields a linear equation for the fluctuation $\delta\rho$.

Linearized TDHF in Frequency Space

After Fourier transform, one obtains

$$(\omega I - \mathcal{L})\,\delta\rho(\omega) = s(\omega),$$

where

- $\mathcal{L}$ is the linearized Liouvillian,
- $s(\omega)$ is the driving term.

This is exactly of the form

$$Ax = b.$$

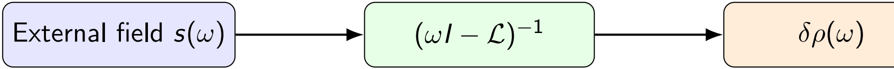
TDHF as an Inverse-Operator Problem

The formal solution is

$$\delta\rho(\omega) = (\omega \boldsymbol{I} - \mathcal{L})^{-1} \boldsymbol{s}(\omega).$$

Thus the TDHF response is a resolvent applied to a source term. This is precisely the same algebraic structure targeted by HHL.

TDHF / Linear Response Diagram



$$\delta\rho(\omega) = (\omega I - \mathcal{L})^{-1} s(\omega)$$

The whole linear-response problem is an inverse-operator map from source to response.

Connection to RPA

In the small-amplitude limit, TDHF is closely related to the RPA eigenvalue problem

$$\begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \omega \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Equivalently, one can write a driven problem

$$(\omega I - M)v = s.$$

Again, the essential structure is resolvent inversion.

Physical Role of QPE in HHL

From a physics viewpoint:

- QPE performs a spectral decomposition of the operator,
- the controlled rotation applies a resolvent-like inverse weight,
- inverse QPE recombines the filtered eigenmodes.

So the logic of HHL is

$$\text{spectral analysis} \rightarrow \text{inverse weighting} \rightarrow \text{state reconstruction.}$$

Unified Conceptual Diagram

$|b\rangle$ → QPE: $|u_j\rangle \to |u_j\rangle|\lambda_j\rangle$ → $\lambda_j \to 1/\lambda_j$ →

This is the same logic as in Green's-function and linear-response theory.

Important Caveats

The conceptual connection is strong, but practical implementation requires:

- efficient Hamiltonian simulation or block encoding,
- well-conditioned matrices,
- efficient state preparation,
- care with non-Hermitian response matrices.

For non-Hermitian problems one often uses a Hermitian embedding such as

$$\mathcal{H} = \begin{pmatrix} 0 & M \\ M^\dagger & 0 \end{pmatrix}.$$

Summary

- In the original HHL algorithm, **QPE is essential**.
- QPE is the block that resolves the operator spectrally.
- HHL then implements the spectral filter

$$\lambda_j \mapsto \frac{1}{\lambda_j}.$$

- This makes HHL naturally connected to
  - resolvents,
  - Green's functions,
  - susceptibilities,
  - TDHF and RPA,
  - frequency-domain linear response.

Final Conceptual Equivalence

$$\boxed{A^{-1}|b\rangle \;\sim\; (zI - H)^{-1}|b\rangle \;\sim\; (\omega I - \mathcal{L})^{-1}s}$$

$$\boxed{\text{HHL} = \text{QPE} + \text{spectral inversion} + \text{state reconstruction}}$$

HHL is therefore best interpreted as a quantum inverse-operator method.